

Daniele Della Cioppa

Android Software Engineer — Secure MDM & Endpoint Control
London, UK



Android Software Engineer focused on Kotlin, secure Android mobile applications, MDM and endpoint control for Ministry of Defence (MOD) environments. I build Android Device Owner and Android Enterprise workflows using Device Policy Manager, kiosk lockdown, controlled APK update delivery, REST API integration, Gradle builds and policy-driven device behaviour.

I also bring identity and infrastructure automation experience across Active Directory, Kerberos/SPNEGO, SMB storage, Group Policy, Folder Redirection and Joiner/Mover/Leaver (JML) workflows using Python and PowerShell, supporting secure mobile estate integration with enterprise Windows and MOD networks.

Work Experience

2025–2026
Current

Software Developer — Akhter Computers

Secure Android, MDM and Infrastructure Automation

- Built secure MOD-environment prototypes combining endpoint-control requirements with Active Directory onboarding, Kerberos/SPNEGO authentication, backend token services and protected file access workflows.
- Implemented a Windows Single Sign-On flow where a domain user obtains Kerberos tickets, negotiates with a Python FastAPI identity service, receives an RS256 JWT, and accesses backend services through JWKS-validated trust.
- Validated Samba Active Directory and Windows Server 2022 lab environments covering AD, DNS, Kerberos, domain logon, Windows 11 Pro domain join behaviour, SPNs, keytabs and GSSAPI service validation.
- Moved the project direction toward enterprise-native AD/SMB/GPO automation: central user storage through SMB, Folder Redirection through Group Policy, JML workflows and Python/PowerShell orchestration.
- Built a Pairing-Based Domain Onboarding Tool for guided server/client onboarding, including pairing flow, client DNS configuration, domain join workflow and validation-oriented state-machine planning.
- Created deployment workflows for Python services using Debian packages, virtual environments, systemd services, environment files, hardened runtime configuration and KVM-backed reproducible lab testing.

2024–2025

Android System Developer — Distributed Device Systems

- Developed Kotlin Android features for distributed digital-signage and device-synchronisation systems using a local-first architecture for secure or limited-connectivity environments.
- Integrated Android clients with REST APIs and Linux-backed services for content delivery, command/status reporting, update orchestration and remote operational tooling.
- Structured Android components around MVVM and Clean Architecture boundaries so UI, device orchestration, API, persistence and policy logic remained maintainable and testable.
- Used Kotlin coroutines and Flow for asynchronous device state, background sync, network calls, live status updates and resilience during intermittent connectivity.

2023–2024

Android System Developer — Remote MDM & Update Infrastructure

- Developed Android MDM and endpoint-control infrastructure for managed devices, including Device Owner provisioning, Android Enterprise policy flows, Device Policy Manager integrations and kiosk mode lockdown.
- Built Kotlin secure Android mobile application components for remote management, device status reporting, controlled command handling and local policy enforcement.
- Implemented secure APK delivery and update workflows with signature/hash verification, controlled rollout behaviour and administrator-authorized privileged actions.
- Worked with Gradle-based Android builds, CI/CD release packaging and automated validation workflows across Android clients, Linux-backed services and operational tooling.

2021–2023

Software Development Apprenticeship

- Contributed to IoT and Android projects involving telemetry, REST/gateway communication, PostgreSQL-backed monitoring, field diagnostics and automated testing.
- Built a foundation in systems development, debugging and customer-facing operational support.

SKILLS

Android

Kotlin / Android SDK
Device Owner
Kiosk Mode
REST APIs

MDM

Android Enterprise
Device Policy Manager
Secure APK Updates
Secure Mobile Apps

Architecture

MVVM
Clean Architecture
Coroutines / Flow
Offline-first Systems

Delivery

Gradle
CI/CD
Automated Testing
Python / PowerShell

Secure Infra

MOD Environments
Active Directory
Kerberos / SPNEGO
SMB / GPO / JML

Main Projects



Android MDM, Device Owner & Kiosk Mode

Developed secure Android endpoint-control features around Android Enterprise Device Owner provisioning, Device Policy Manager policy enforcement, kiosk lockdown, remote management, secure APK updates and administrator-authorised privileged actions.

Kotlin / MDM



Distributed Android Digital Signage Platform

Worked on Kotlin Android-based digital-signage and synchronisation systems focused on REST API integration, content delivery, remote device management, Gradle release packaging, update distribution and offline-first resilience.

REST / Gradle



Pairing-Based Domain Onboarding Platform

Built a guided onboarding platform for Windows Server and Active Directory environments, covering server/client pairing, DNS configuration, domain-join validation, SMB home-directory setup, Group Policy and Folder Redirection workflows.

AD / GPO



Windows SSO / Kerberos to JWT Prototype

Designed a working authentication flow where Windows domain login is reused to access protected backend services through Kerberos/SPNEGO validation and RS256 JWT trust.

SSO